



A

Assessment Report
on
“Credit Card Fraud”
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By

Vanshika Aggarwal (202401100400207)

Under the supervision of

“Abhishek Shukla”

KIET Group of Institutions, Ghaziabad

An Autonomous Institute affiliated to

Dr. A.P.J. Abdul Kalam Technical University, Lucknow
(Formerly UPTU)

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Introduction

With the rapid growth of online and card-based payments, credit card fraud has become a major concern for financial institutions. It can lead to significant financial losses, reduce customer trust, and harm the reputation of banks and payment providers. Fraud can occur in various forms, including identity theft, stolen card information, and system manipulation.

Manual methods of detecting fraud are often slow and ineffective, especially as fraud techniques become more advanced. To overcome this, machine learning-based systems are now widely used to automatically analyze large amounts of transaction data and detect unusual or suspicious patterns in real-time.

The goal of this project is to build a machine learning model to detect fraudulent credit card transactions. We use a dataset containing both normal and fraud cases, with features like transaction amount, time, and anonymized behavior patterns.

We apply a **Random Forest Classifier**, which uses multiple decision trees to improve prediction accuracy. The data is split into training and testing sets using **stratified sampling** to maintain the balance of fraud and non-fraud cases.

The model classifies transactions into:

- **Class 0:** Legitimate
- **Class 1:** Fraudulent

We evaluate performance using **accuracy**, **precision**, **recall**, **F1-score**, and a **confusion matrix**. Because fraud cases are rare, we focus more on precision and recall than accuracy.

This project shows how machine learning can help detect fraud and lays the foundation for more advanced techniques in the future.



Methodology

The goal of this project is to build a machine learning model to detect fraudulent credit card transactions. The steps followed in this project are as follows:

1. **Data Collection**

The dataset containing historical transaction records was uploaded into Google Colab. It includes various anonymized features related to user behavior, transaction details, and a `Class` label indicating whether a transaction is fraudulent (1) or not (0).

2. **Data Exploration and Preprocessing**

Basic data inspection was performed using `.shape`, `.head()`, and `value_counts()` to understand the structure and distribution of the data. The target variable (`Class`) was found to be highly imbalanced, which is common in fraud detection problems.

3. **Feature and Target Separation**

The features (`x`) and the target label (`y`) were separated. The target column `Class` was removed from the features using `drop()`.

4. **Train-Test Split**

The dataset was split into training and testing sets using an 80-20 ratio, with `stratify=y` to ensure the class distribution is maintained in both subsets.

5. **Model Training**

A **Random Forest Classifier** was chosen for this classification task due to its robustness and ability to handle imbalanced data reasonably well. The model was trained using 100 decision trees (`n_estimators=100`).

6. **Prediction and Evaluation**

Predictions were made on the test set. Model performance was evaluated using metrics such as **accuracy**, **classification report** (which includes precision, recall, and F1-score), and a **confusion matrix** heatmap to visualize the model's classification capability.

Typed Code

```
# STEP 1: INSTALL & IMPORT LIBRARIES
!pip install -q matplotlib seaborn scikit-learn

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, classification_report,
accuracy_score

# STEP 2: LOAD THE DATA (UPLOAD IN COLAB)
from google.colab import files
uploaded = files.upload()

# After upload, read the file (adjust filename if needed)
df = pd.read_csv("7.Predict Credit Card Fraud.csv")

# STEP 3: BASIC INFO
print("Shape:", df.shape)
print(df["Class"].value_counts()) # 0 = normal, 1 = fraud
df.head()

# STEP 4: SEPARATE FEATURES AND TARGET
X = df.drop("Class", axis=1)
y = df["Class"]

# STEP 5: TRAIN-TEST SPLIT
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42, stratify=y)

# STEP 6: TRAIN RANDOM FOREST CLASSIFIER
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# STEP 7: MAKE PREDICTIONS
y_pred = model.predict(X_test)

# STEP 8: EVALUATE MODEL
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
# STEP 9: PLOT CONFUSION MATRIX
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=["Not
Fraud", "Fraud"], yticklabels=["Not Fraud", "Fraud"])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

Output

Choose Files

7. Predict C...d Fraud.csv

• 7. Predict Credit Card Fraud.csv(text/csv) - 150828752 bytes, last modified: 4/18/2025 - 100% done

Saving 7. Predict Credit Card Fraud.csv to 7. Predict Credit Card Fraud (3).csv

Shape: (284807, 31)

Class

0 284315

1 492

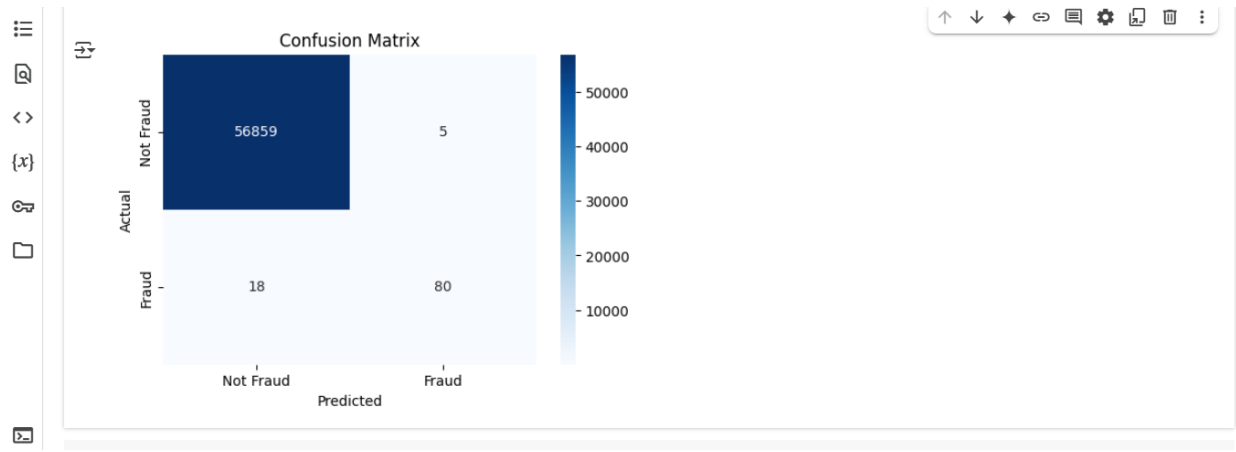
Name: count, dtype: int64

Accuracy: 0.9995962220427653

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	56864
1	0.94	0.82	0.87	98
accuracy			1.00	56962
macro avg	0.97	0.91	0.94	56962
weighted avg	1.00	1.00	1.00	56962

Confusion Matrix



References/Credits

1. Credit Card Fraud Detection Dataset - Kaggle
2. Scikit-learn Documentation - <https://scikit-learn.org>
3. Matplotlib Documentation - <https://matplotlib.org>
4. Seaborn Documentation - <https://seaborn.pydata.org>
5. Random Forest Classifier Explained – GeeksforGeeks
6. Image Used - <https://jupiter.money/blog/credit-card-fraud-detection/>